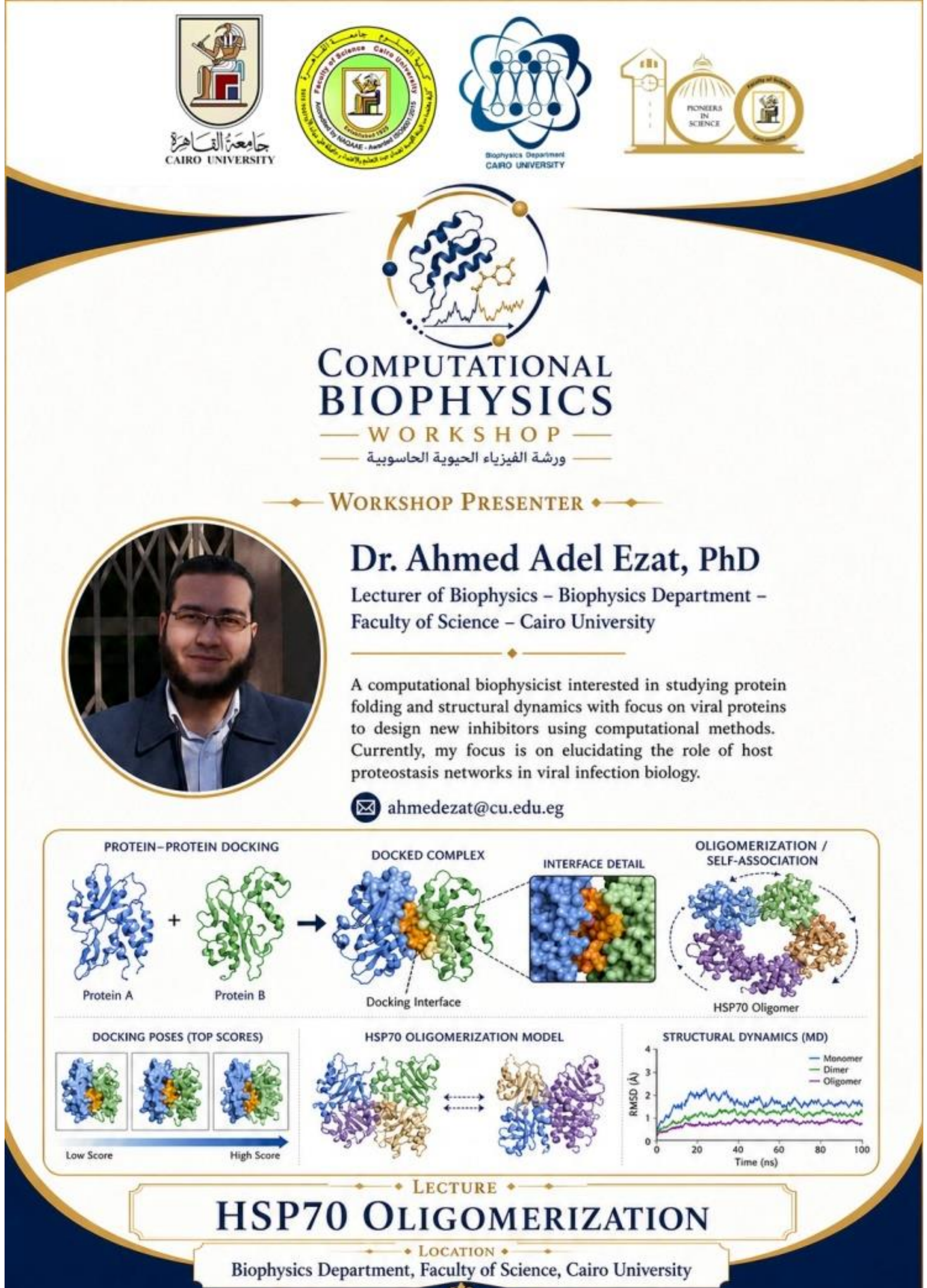
HSP70 OLIGOMERIZATION

A STRUCTURAL PERSPECTIVE

AHMED A. EZAT, PHD.
BIOPHYSICS DEPARTMENT
FACULTY OF SCIENCE
CAIRO UNIVERSITY


جامعة القاهرة
CAIRO UNIVERSITY
Biophysics Department
CAIRO UNIVERSITY
PIONEERS IN SCIENCE
COMPUTATIONAL
BIOPHYSICS
WORKSHOP
ورشة الفيزياء الحيوية الحاسوبية
WORKSHOP PRESENTER
Dr. Ahmed Adel Ezat, PhD
Lecturer of Biophysics – Biophysics Department – Faculty of Science – Cairo University
A computational biophysicist interested in studying protein folding and structural dynamics with focus on viral proteins to design new inhibitors using computational methods. Currently, my focus is on elucidating the role of host proteostasis networks in viral infection biology.
ahmedezat@cu.edu.eg
PROTEIN–PROTEIN DOCKING
Protein A
Protein B
DOCKED COMPLEX
Docking Interface
INTERFACE DETAIL
OLIGOMERIZATION / SELF-ASSOCIATION
HSP70 Oligomer
DOCKING POSES (TOP SCORES)
Low Score
High Score
HSP70 OLIGOMERIZATION MODEL
STRUCTURAL DYNAMICS (MD)
RMSD (Å)
Time (ns)
Monomer
Dimer
Oligomer
LECTURE
HSP70 OLIGOMERIZATION
LOCATION
Biophysics Department, Faculty of Science, Cairo University

STDF

هيئة تمويل العلوم والتكنولوجيا والإبتكار

Science, Technology & Innovation Funding Authority

- STDF funded project "49138" under One Health Call:

"Detection, characterization and inhibition of endoplasmic reticulum (ER) chaperones (GRP78 and GRP94) as biomarkers of several zoonotic diseases caused by viral infections"

OUTLINES

1. HSP70 (STRUCTURE & FUNCTIONS).
2. CONFORMATIONAL CYCLE OF HSP70.
3. OLIGOMERIZATION OF HSP70 (PROPOSED MODELS).
4. CHAPERONE CODE.

COMPUTATIONAL
BIOPHYSICS
WORKSHOP
ورشة الفيزياء الحيوية الحاسوبية

HSP70 "STRUCTURE & FUNCTIONS"

HSP70 ISOFORMS

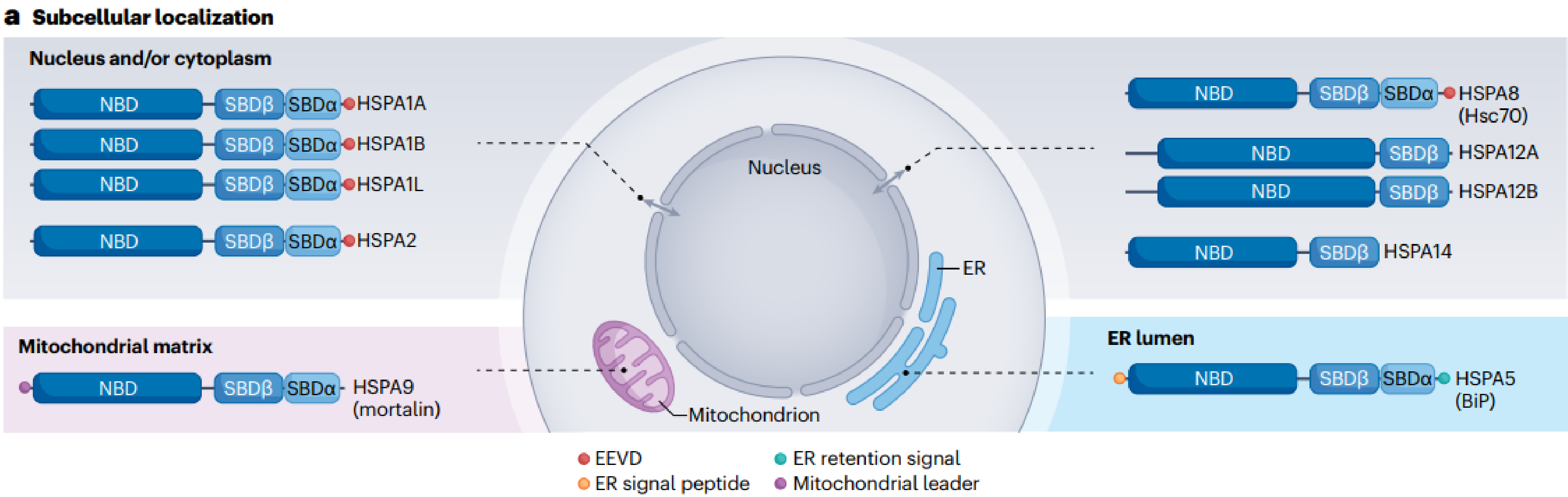
- In humans, there are many isoforms of HSP70.

I. Cytosolic HSP70 (HSPA8/HSC70)

II. ER – resident HSP70 (HSPA5/BiP/GRP78)

III. Mitochondria – resident HSP70 (HSPA9/Mortalin/GRP75)

HSP70 ISOFORMS



Wentink, A., Rosenzweig, R., Kampinga, H. et al. Mechanisms and regulation of the Hsp70 chaperone network. Nat Rev Mol Cell Biol 27, 110–128 (2026).

HSP70 FUNCTIONS

- Hsp70 chaperone is essential to:

I. Maintain cellular protein homeostasis (Quality control of proteins).

II. Facilitating protein folding and prevent misfolding/aggregation. (Housekeeping)

III. Assembly.

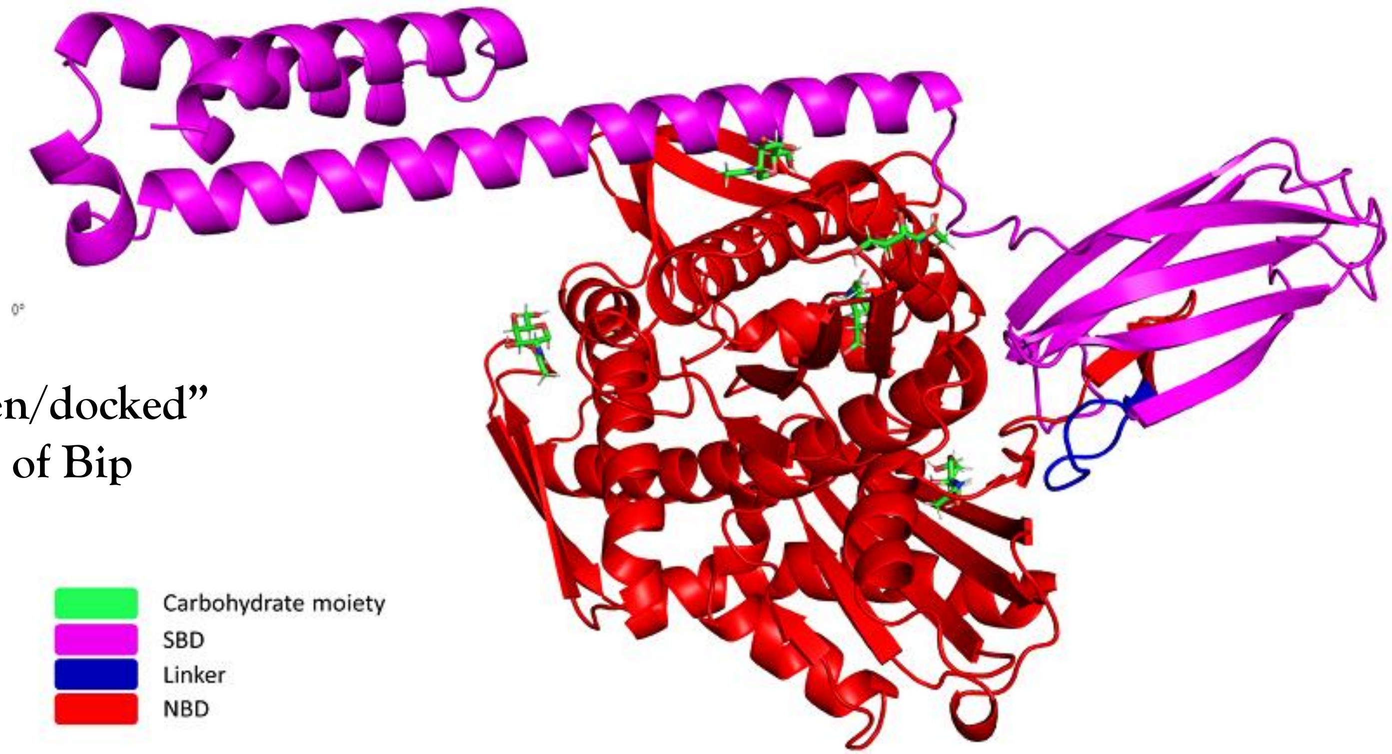
IV. Membrane translocation.

V. Foldase/Holdase.

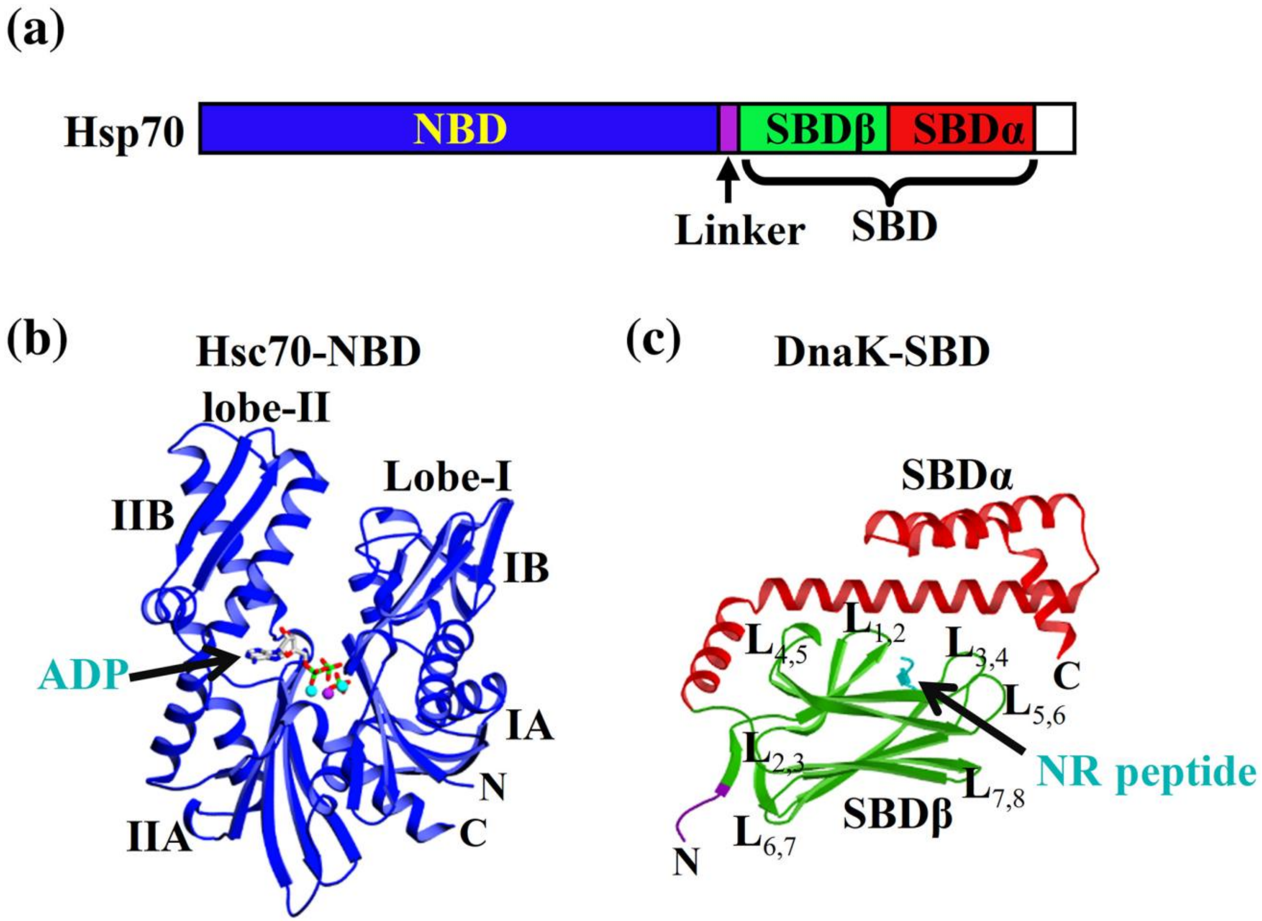
VI. Disaggregase.

HSP70 STRUCTURE

- HSP70 (~ 640 - 650 amino acids) is comprised of two domains, a nucleotide-binding domain (NBD) and a substrate-binding domain (SBD), linked by a flexible hydrophobic linker.



ATP - bound "open/docked" Conformation of Bip



(1HPM)

(1DKX)

Structural and functional analysis of the Hsp70/Hsp40 chaperone system

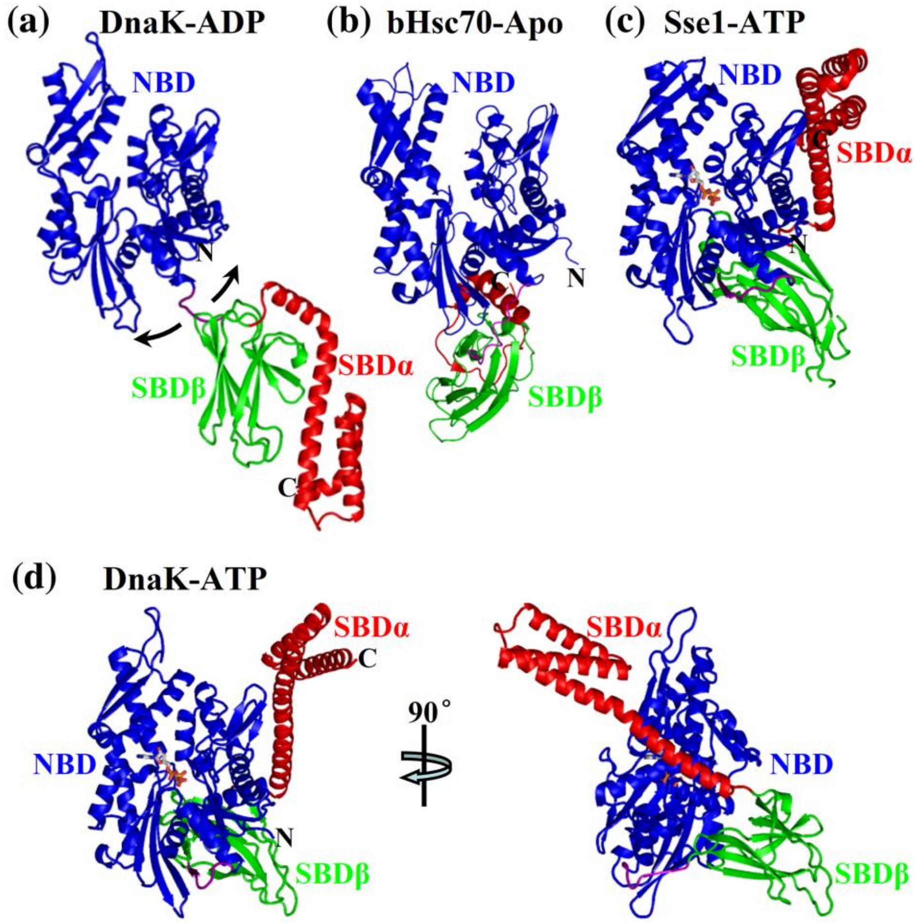
Protein Science, Volume: 29, Issue: 2, Pages: 378-390, First published: 11 September 2019, DOI: (10.1002/pro.3725)

(1YUW)

(2KHO)

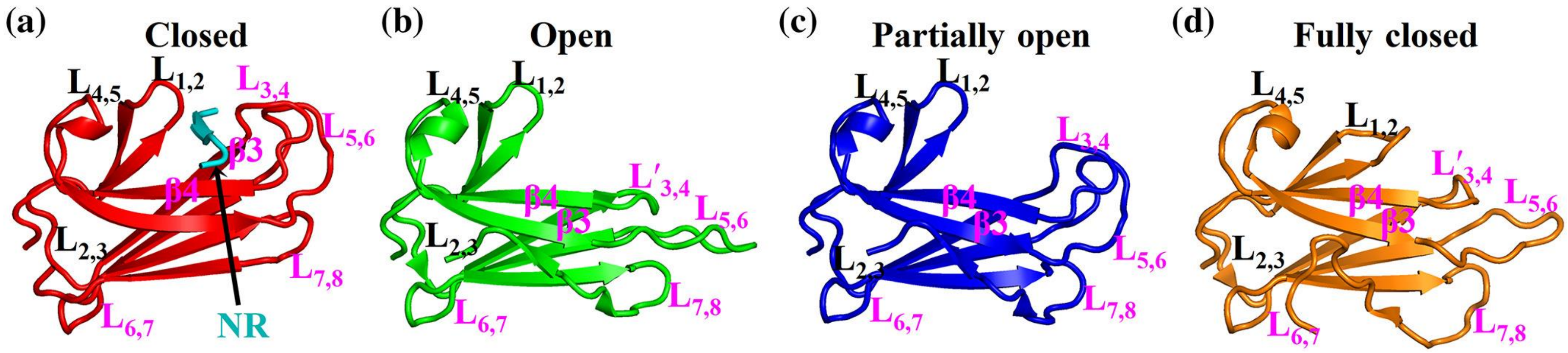
(2QXL)

(4JNE)



Structural and functional analysis of the Hsp70/Hsp40 chaperone system

Protein Science, Volume: 29, Issue: 2, Pages: 378-390, First published: 11 September 2019, DOI: (10.1002/pro.3725)

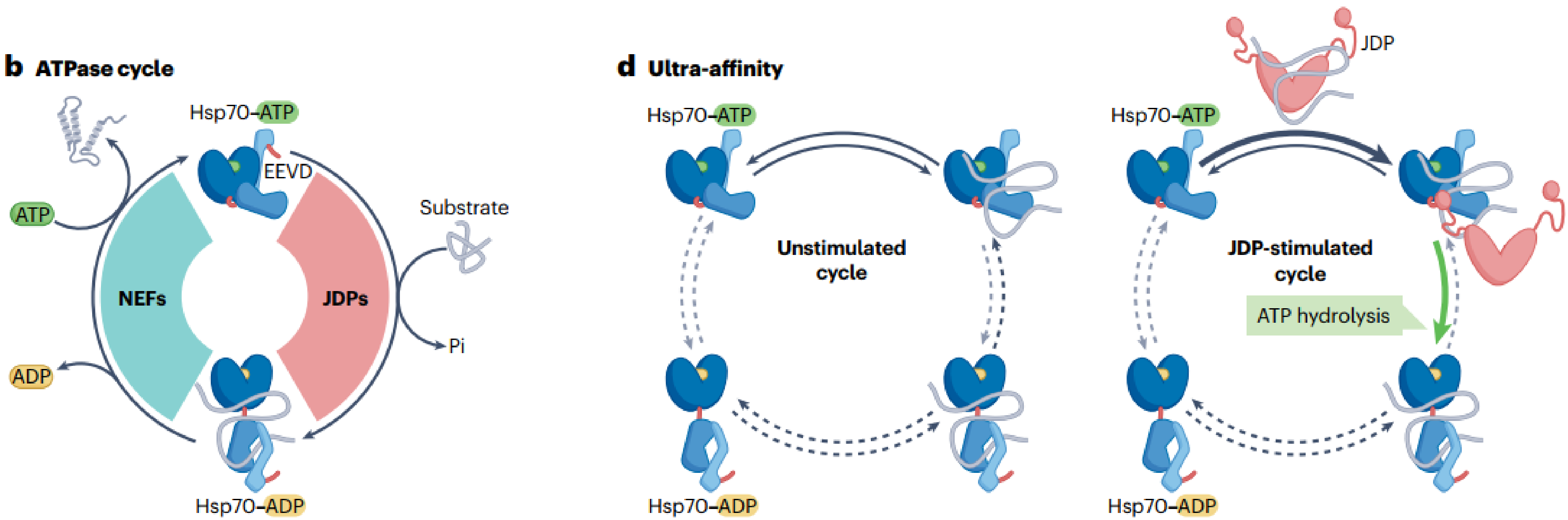


(1DKX) DnaK - ADP	(4JNE) DnaK - ATP	(4B9Q) DnaK - ATP	(6ASY) BiP - ATP

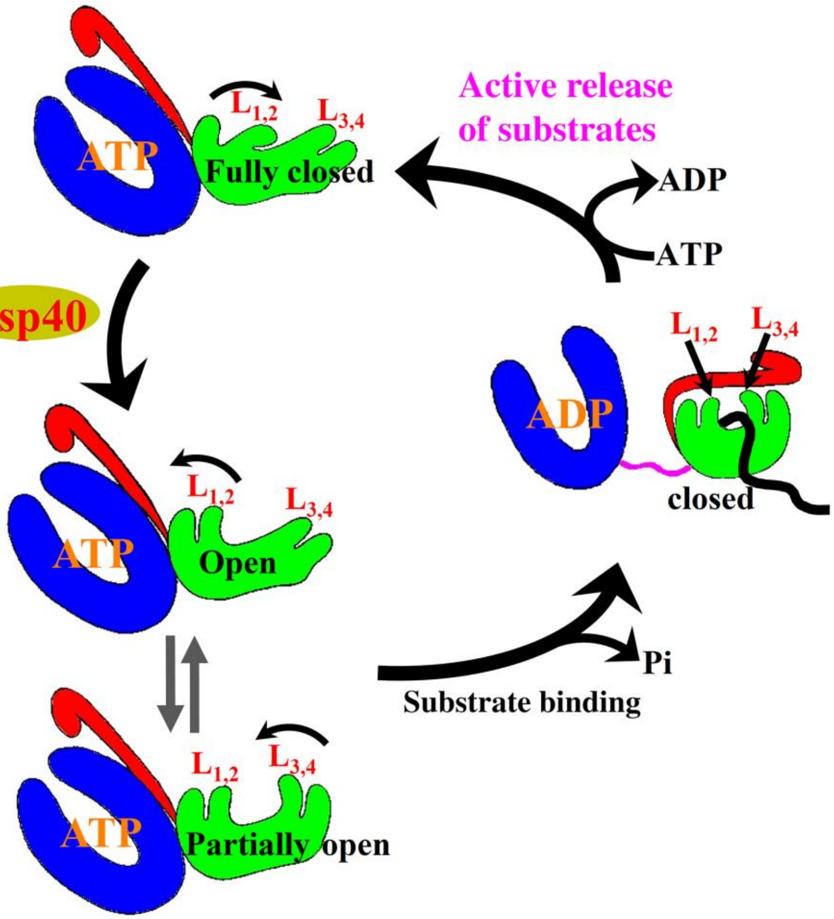
Conformations of the polypeptide-binding pocket. “The SBDβ structures”

Structural and functional analysis of the Hsp70/Hsp40 chaperone system
Protein Science, Volume: 29, Issue: 2, Pages: 378-390, First published: 11 September 2019, DOI: (10.1002/pro.3725)

Conformational Cycle of HSP70



Wentink, A., Rosenzweig, R., Kampinga, H. et al. Mechanisms and regulation of the Hsp70 chaperone network. Nat Rev Mol Cell Biol 27, 110–128 (2026).



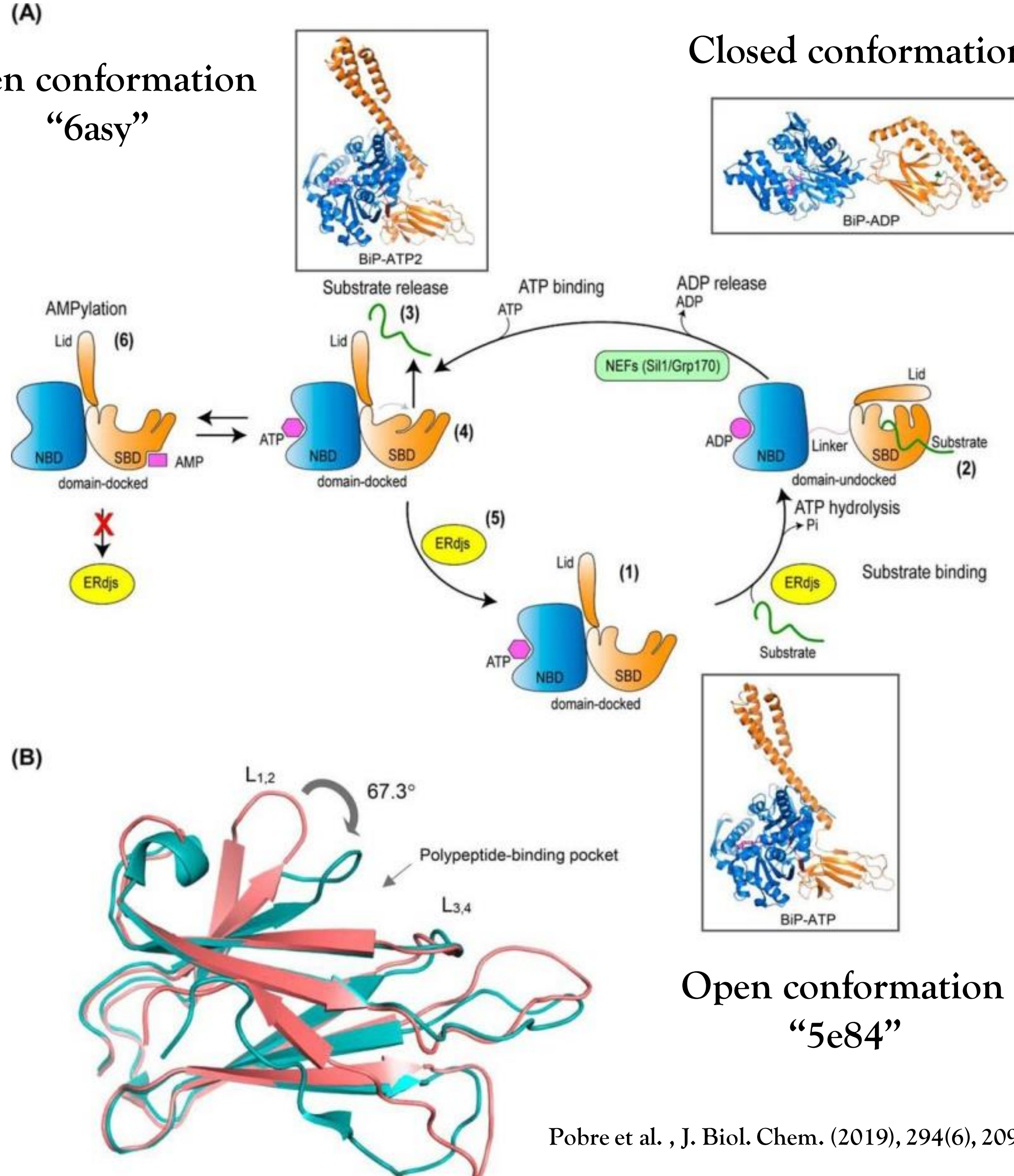
Schematic of the proposed Hsp70 chaperone cycle.

Structural and functional analysis of the Hsp70/Hsp40 chaperone system
Protein Science, Volume: 29, Issue: 2, Pages: 378-390, First published: 11 September 2019, DOI: (10.1002/pro.3725)

BiP ATPase cycle

Open conformation
"6asy"

Closed conformation

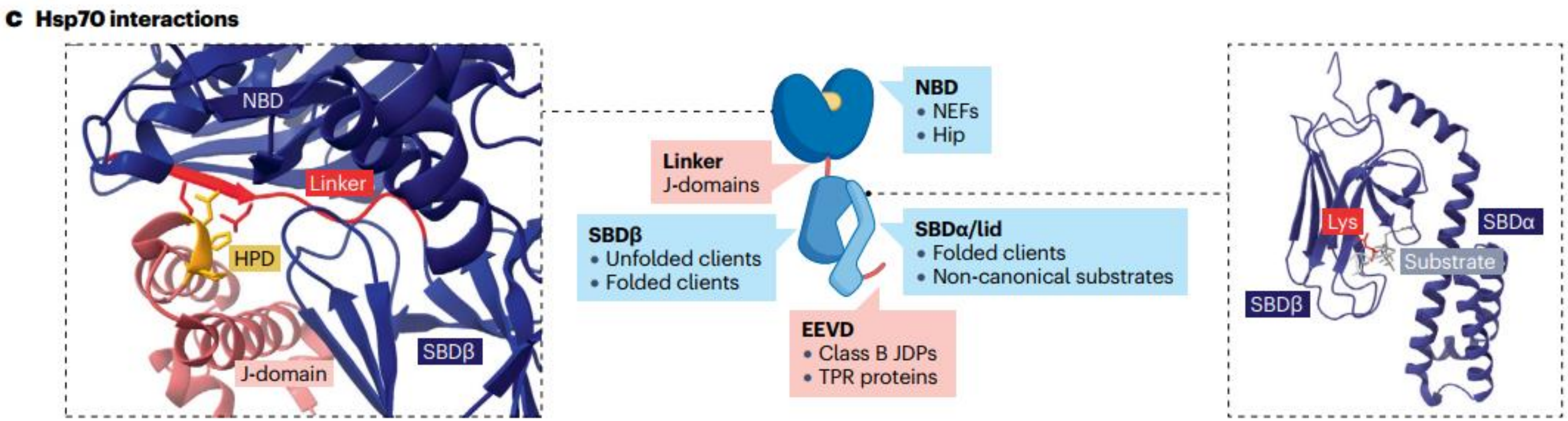


Open conformation
"5e84"

Pobre et al. , J. Biol. Chem. (2019), 294(6), 2098-2108

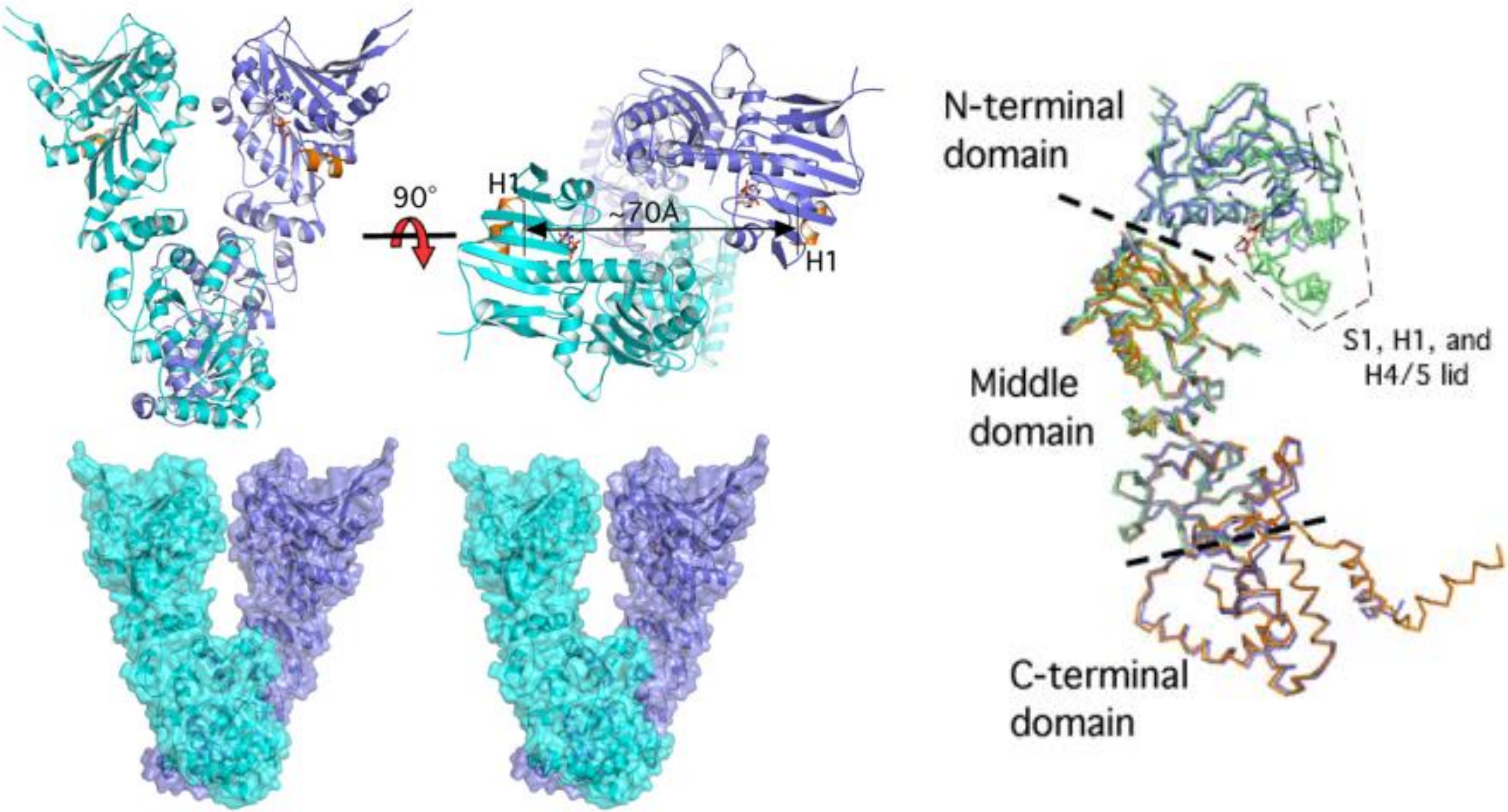
HSP70 INTERACTIONS WITH OTHER HEAT SHOCK PROTEINS

- HSP70 interacts with other Heat Shock Proteins like HSP90, HSP40 and HSP110.
- The interaction is mediated by NBD or SBD.



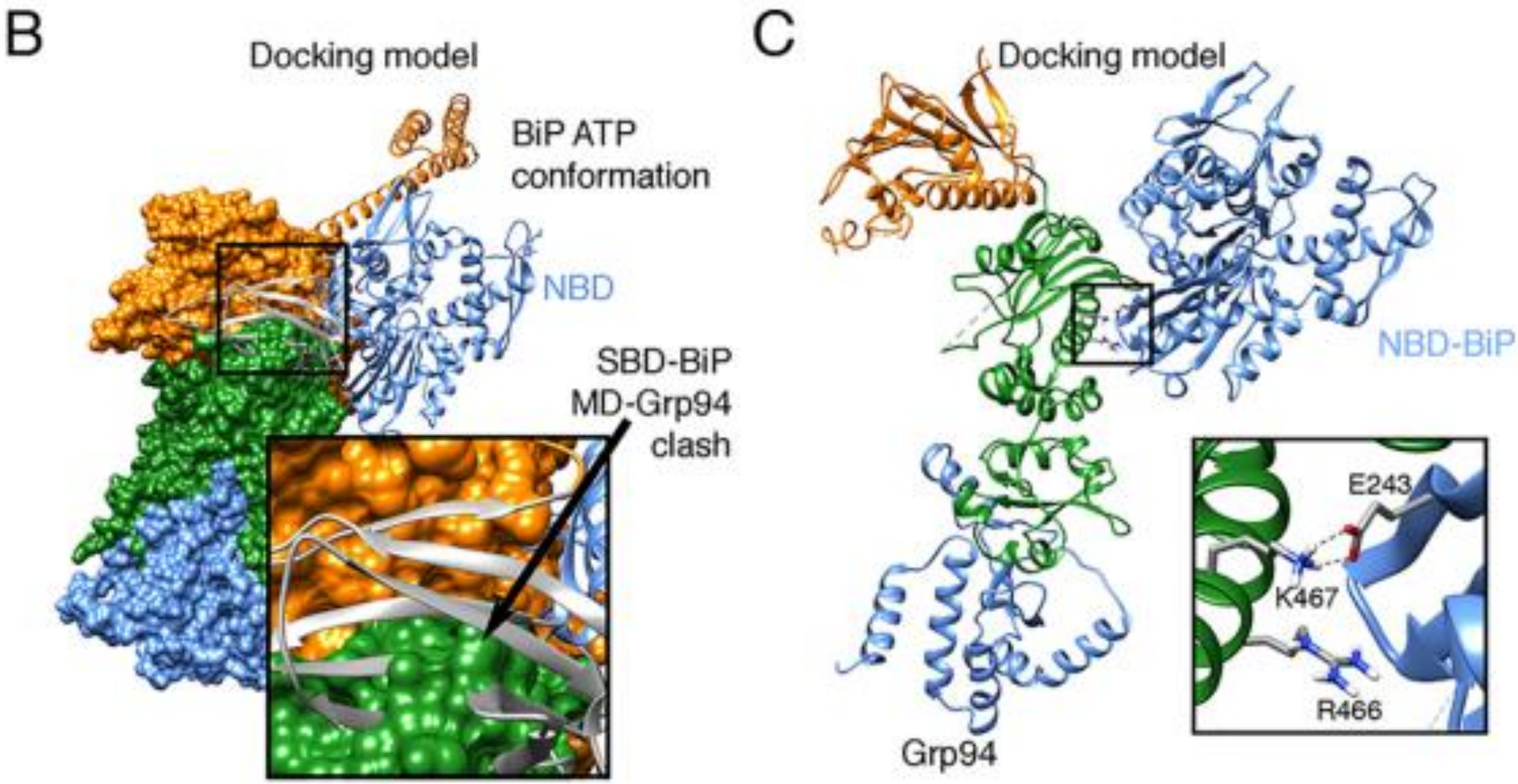
Wentink, A., Rosenzweig, R., Kampinga, H. et al. Mechanisms and regulation of the Hsp70 chaperone network. Nat Rev Mol Cell Biol 27, 110–128 (2026).

HSP90

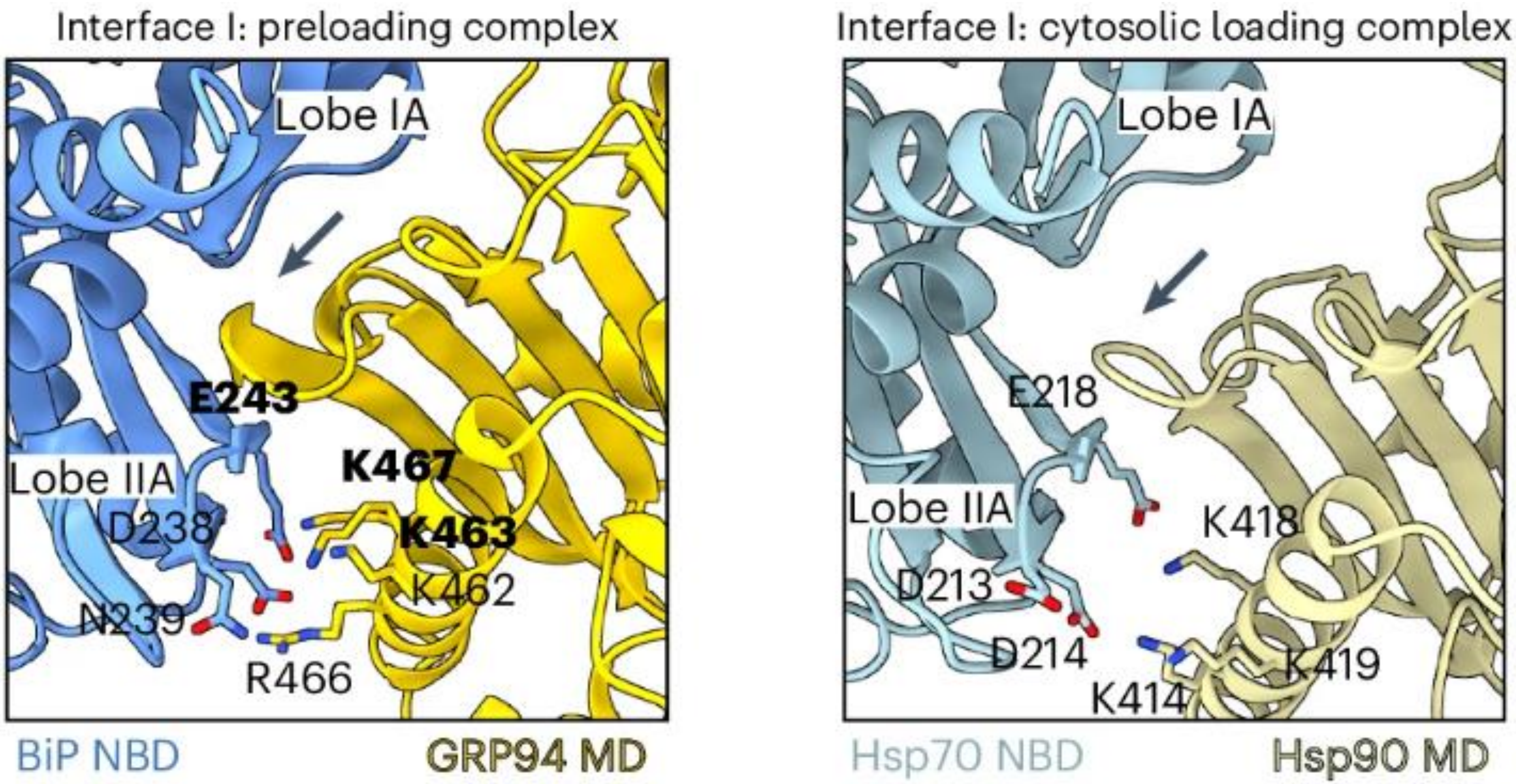


Dollins et al. , Molecular Cell, (2007), 28, 41–56

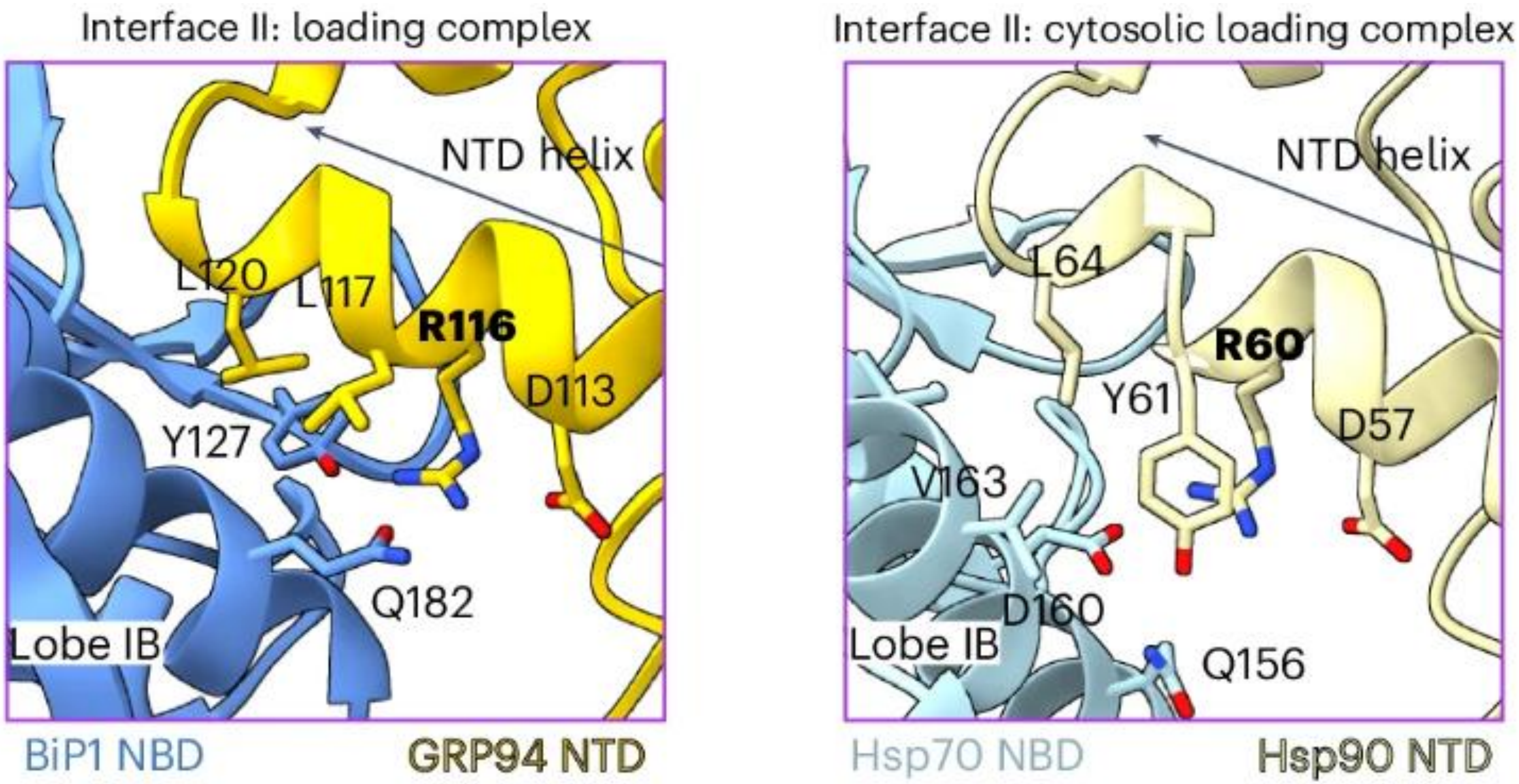
HSP70 – HSP90 Complex



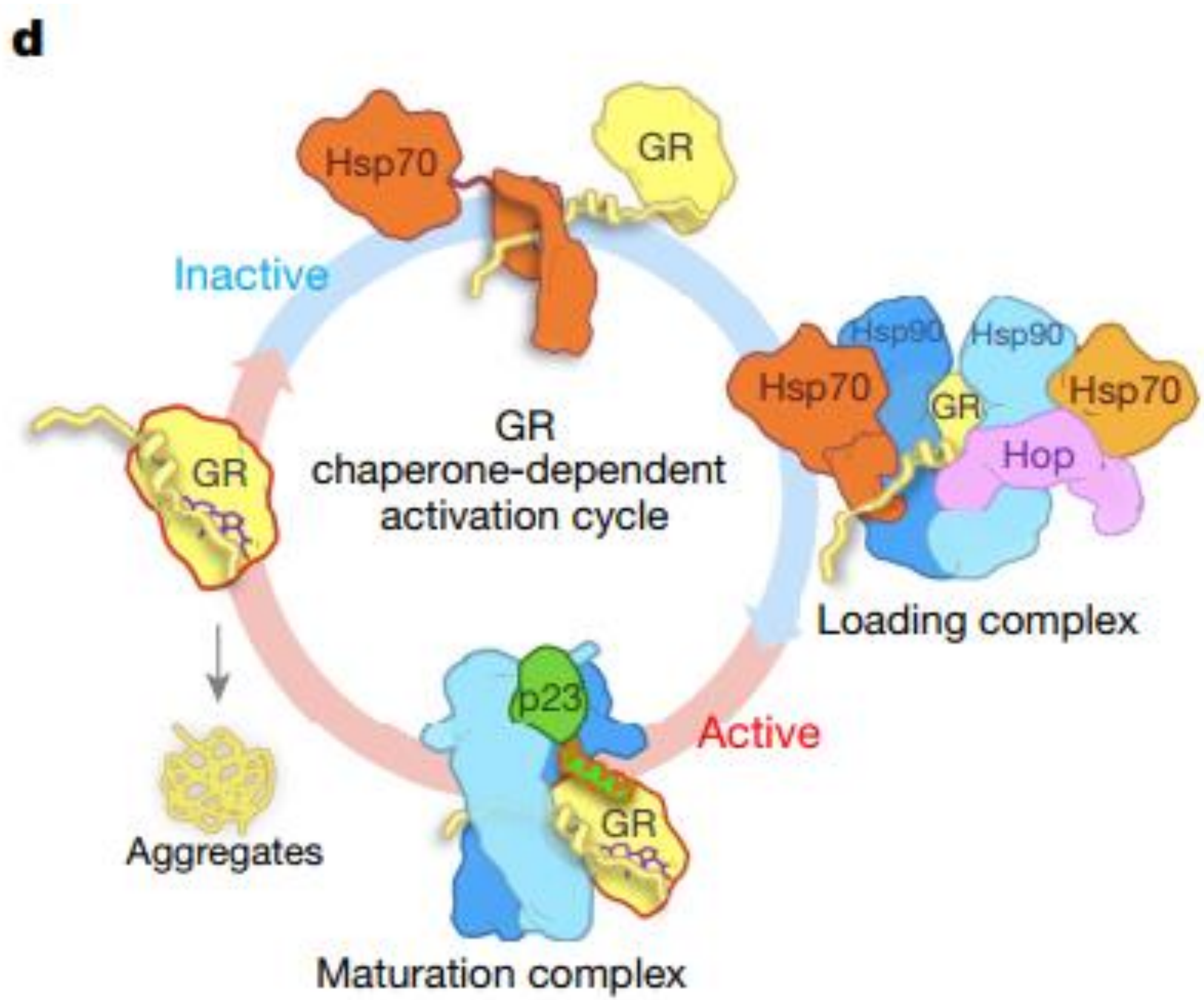
Sun et al. , J. Biol. Chem. (2019), 294(16) 6387–6396



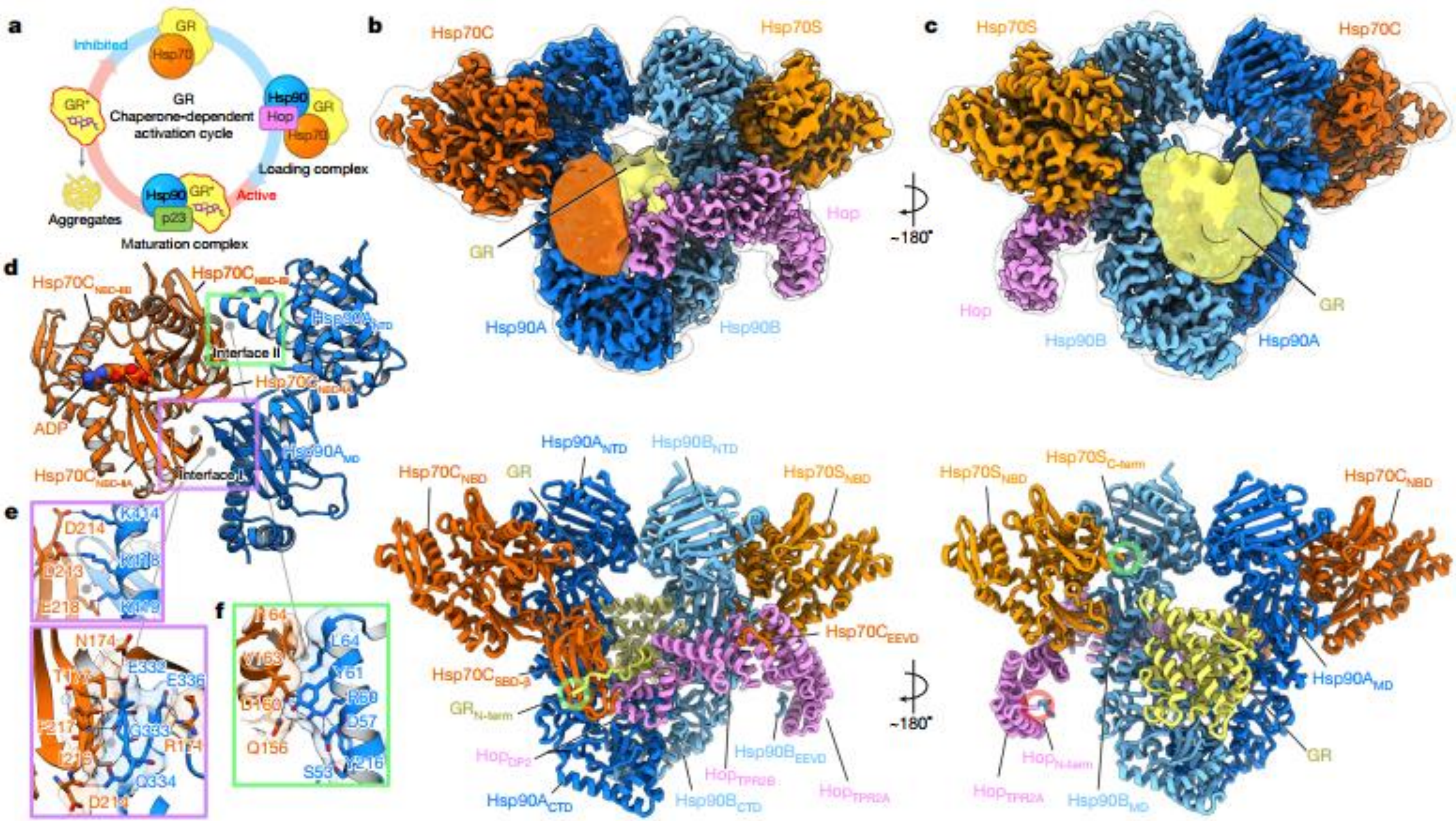
Brenner, J.C., Zirden, L.C., Buzuk, L. et al. Conformational plasticity of a BiP–GRP94 chaperone complex.
Nat Struct Mol Biol 32, 1947–1958 (2025).



Brenner, J.C., Zirden, L.C., Buzuk, L. et al. Conformational plasticity of a BiP–GRP94 chaperone complex. Nat Struct Mol Biol 32, 1947–1958 (2025).

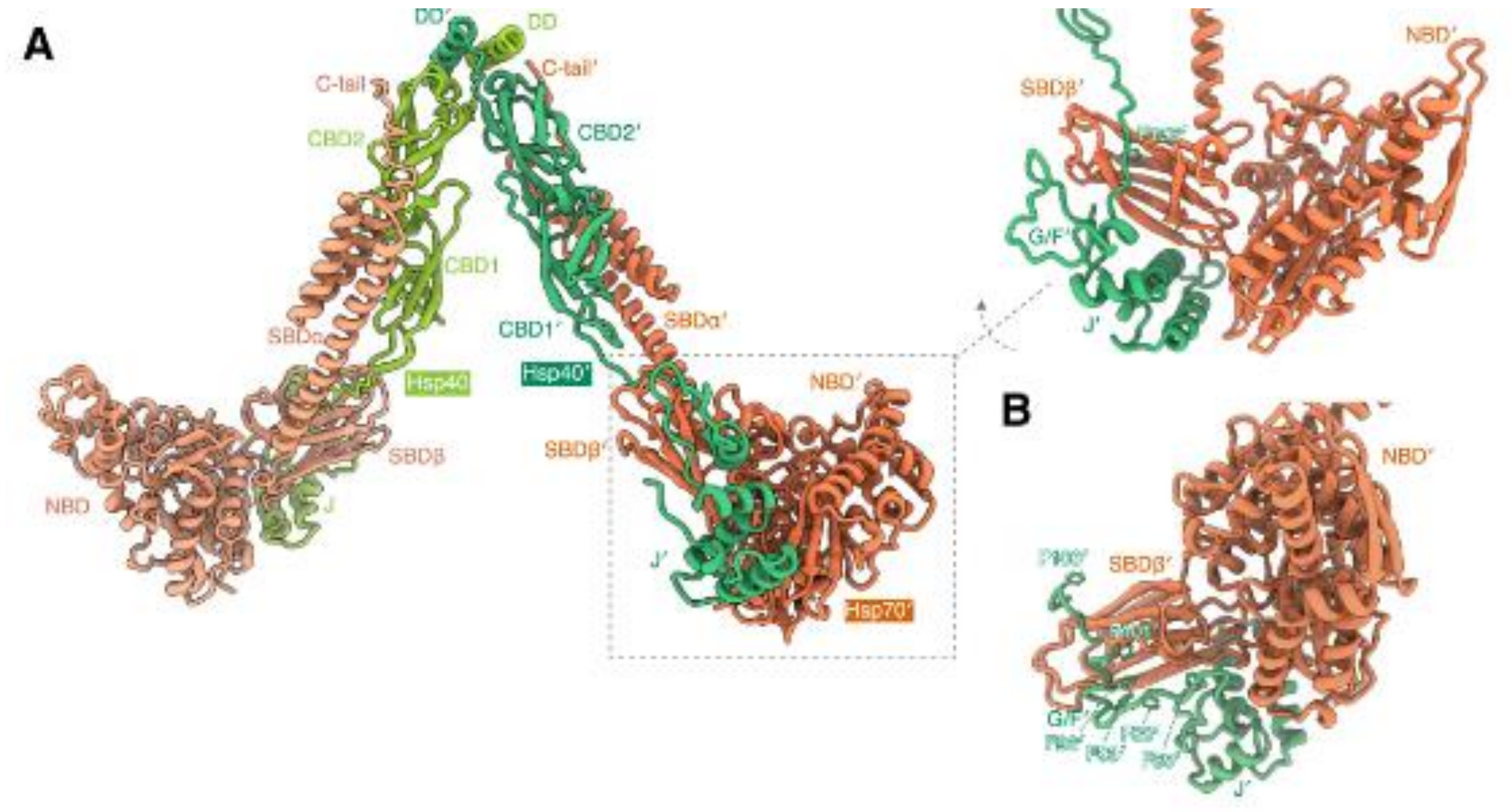


Wang et al. , Nature (2022), Vol.601



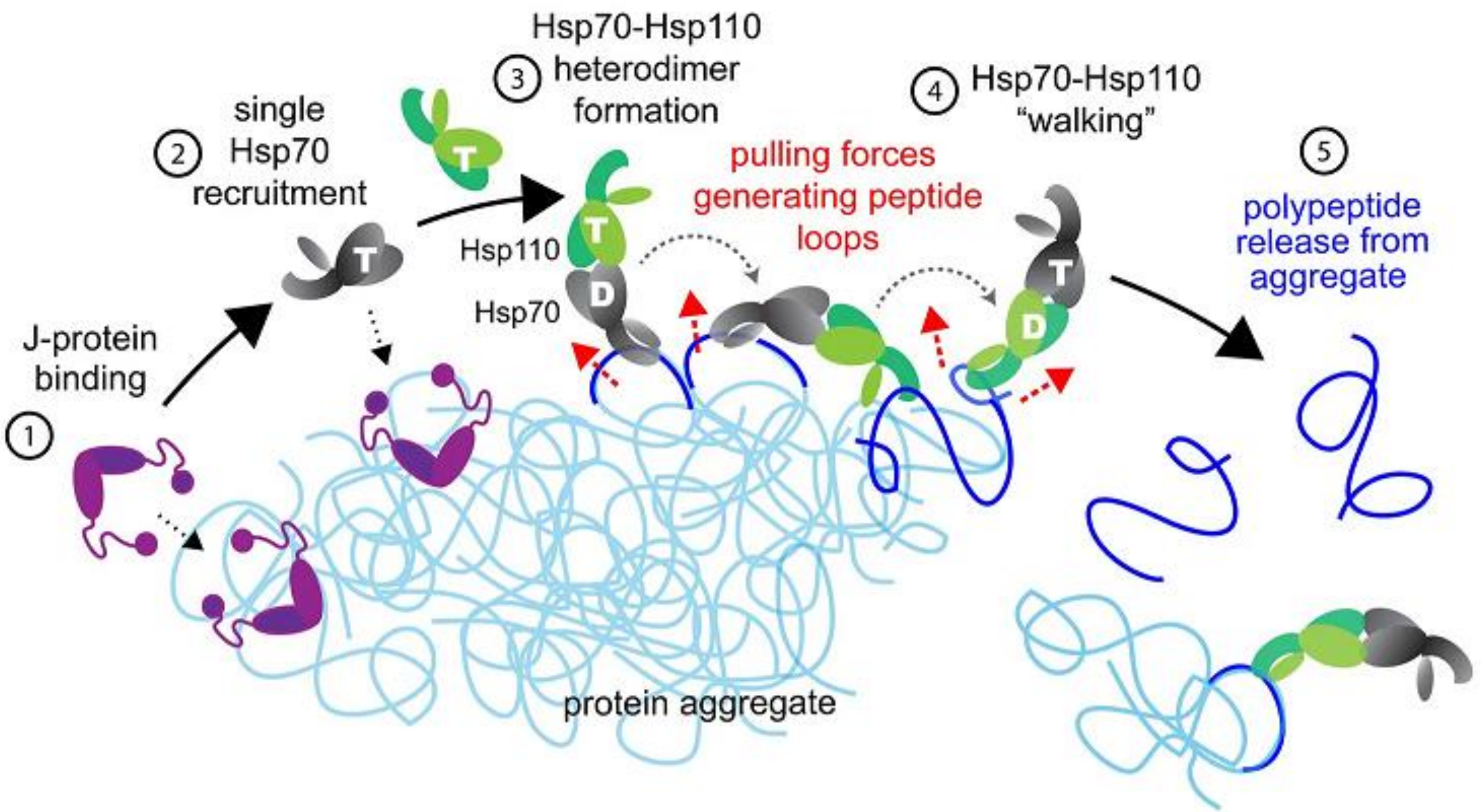
Wang et al. , Nature (2022), Vol.601

HSP70 – HSP40 Complex



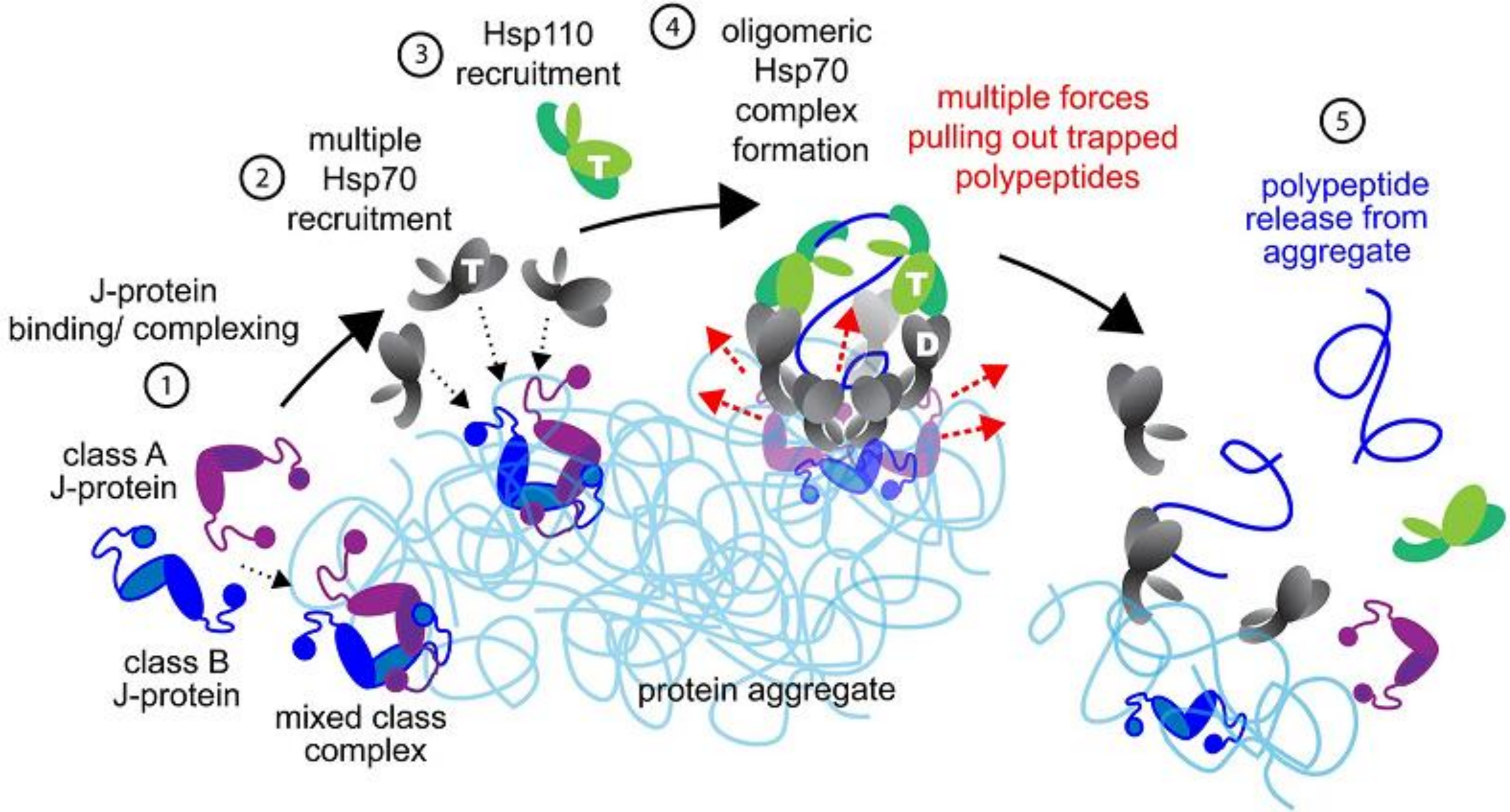
Mechanisms of assembly and function of the Hsp70-Hsp40 chaperone machinery, Molecular Cell, 2025

HSP70 Disaggregase Function



Metazoan Hsp70-based protein disaggregases: emergence and mechanisms, Front. Mol. Biosci., 2015

HSP70 Disaggregase Function

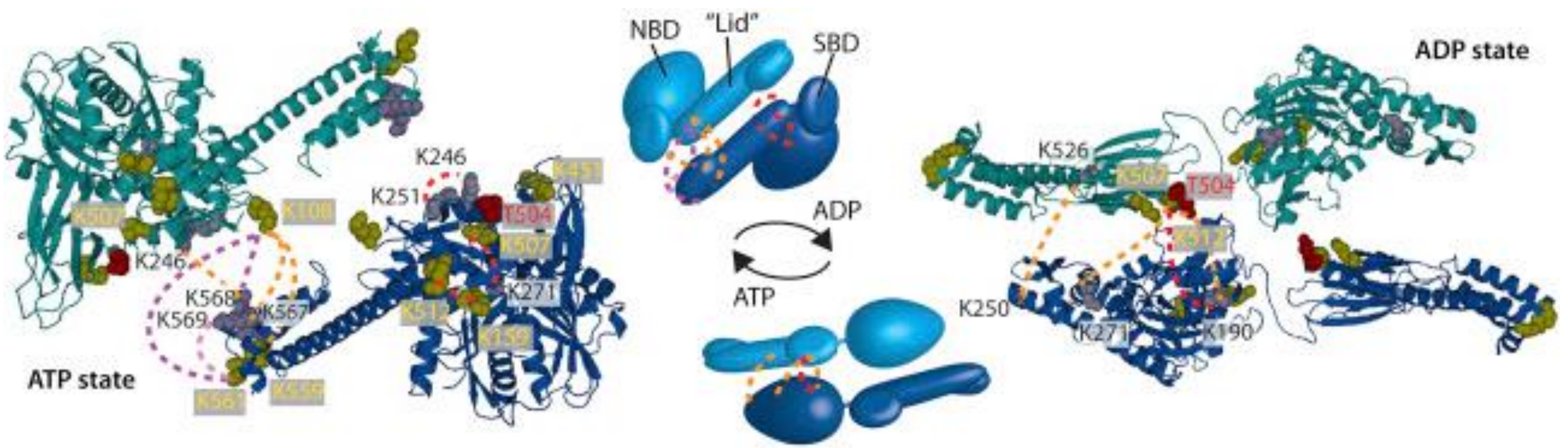


Metazoan Hsp70-based protein disaggregases: emergence and mechanisms, Front. Mol. Biosci., 2015

HSP70 OLIGOMERIZATION

HSP70 DIMERIZATION

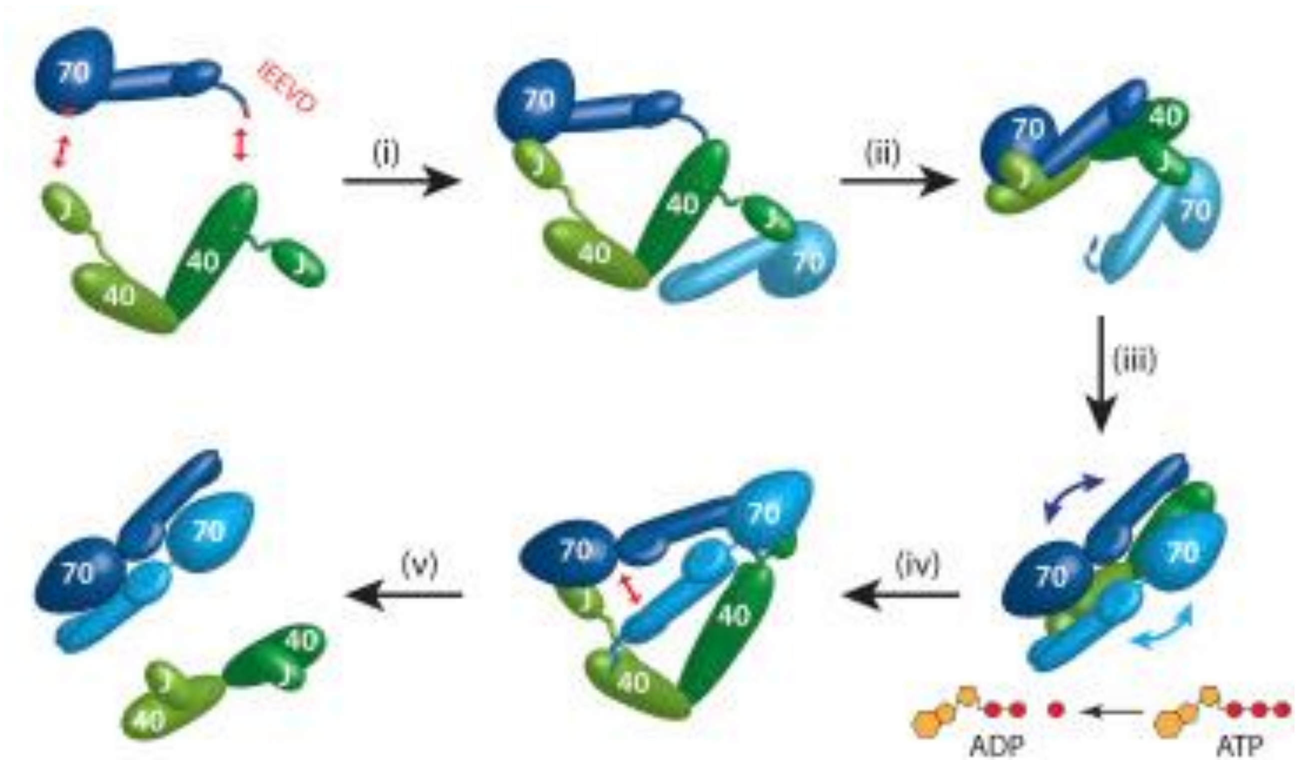
Hsp70 Forms Antiparallel Dimers



Hsp70 Forms Antiparallel Dimers Stabilized by Post-translational Modifications to Position Clients for Transfer to Hsp90, Cell Reports, Volume 11, Issue 5, 5 May 2015, Pages 759-769

HSP70 DIMERIZATION

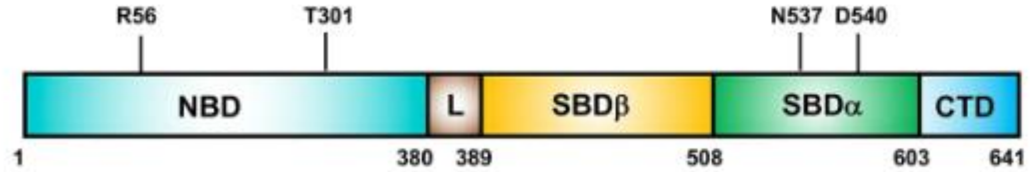
Hsp70 Forms Antiparallel Dimers

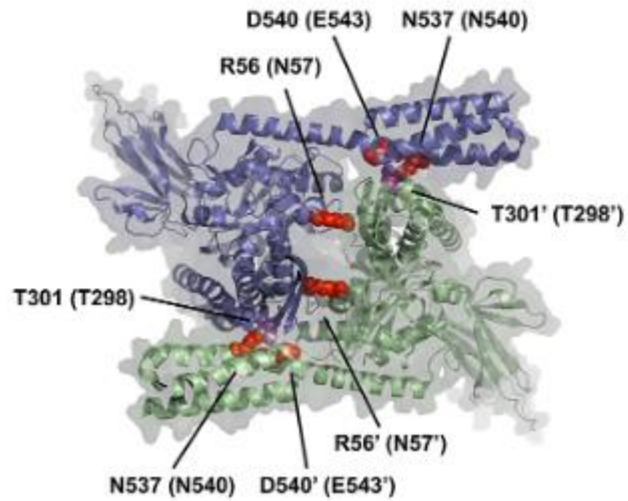


Hsp70 Forms Antiparallel Dimers Stabilized by Post-translational Modifications to Position Clients for Transfer to Hsp90, Cell Reports, Volume 11, Issue 5, 5 May 2015, Pages 759-769

HSP70 DIMERIZATION

Hsp70 Forms Antiparallel Dimers

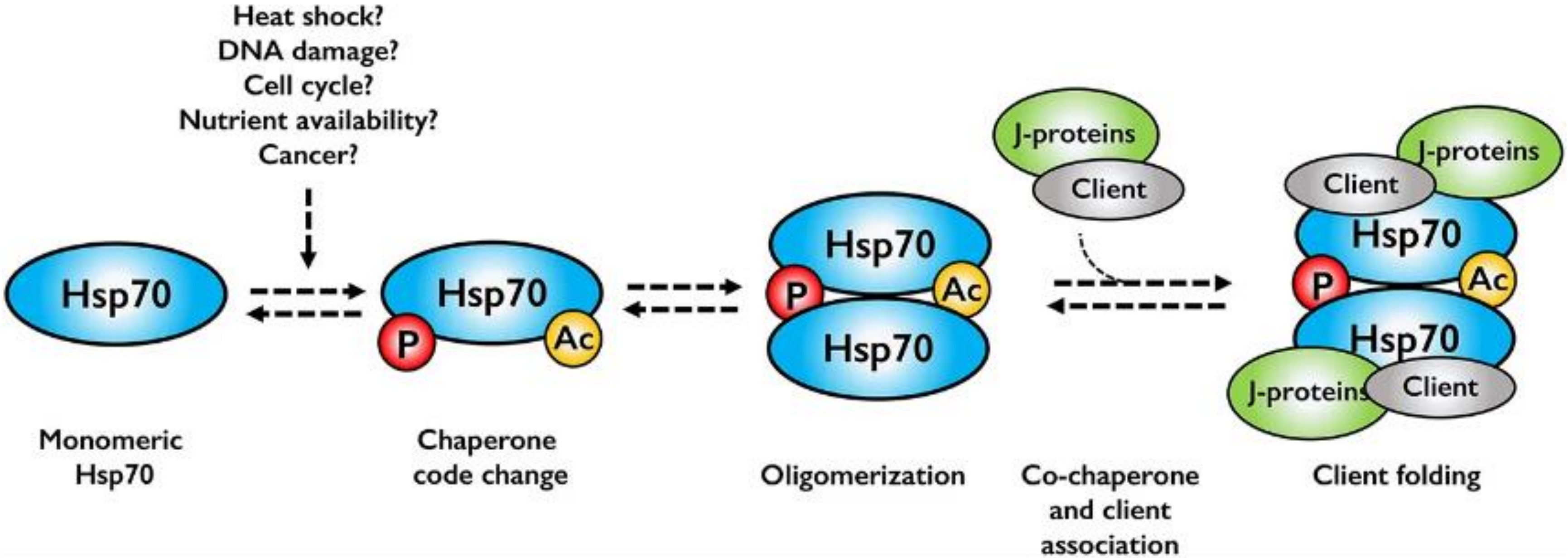




Oligomerization of Hsp70: Current Perspectives on Regulation and Function, Front. Mol. Biosci., 04 September 2019

HSP70 OLIGOMERIZATION

Higher oligomerization of Hsp70 is facilitated and stabilized by PTMs

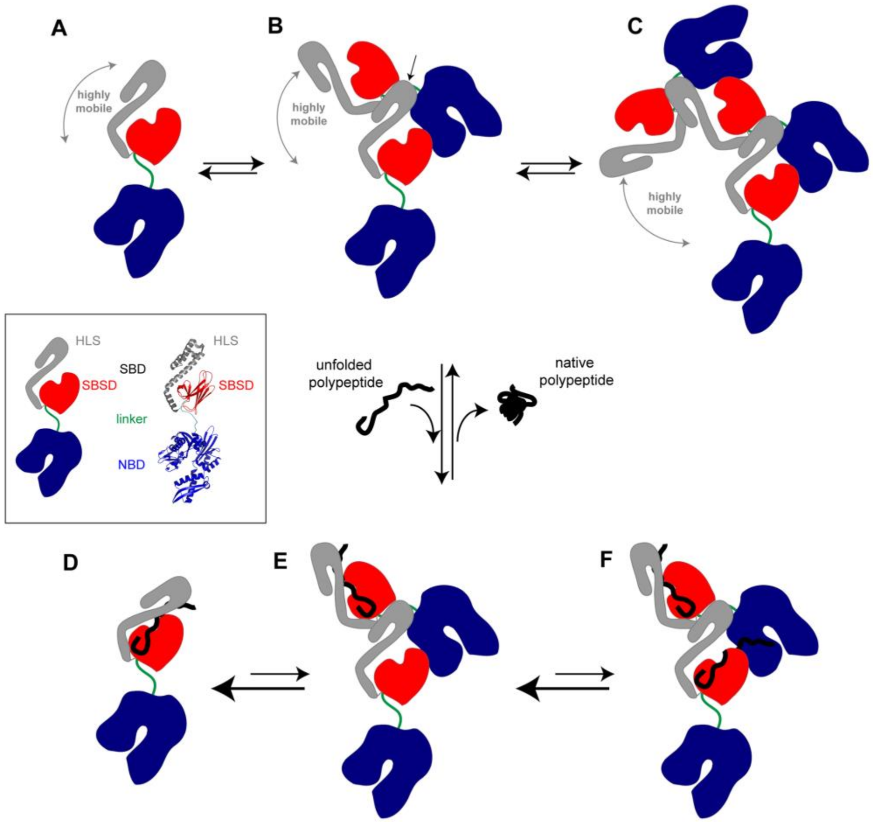


Oligomerization of Hsp70: Current Perspectives on Regulation and Function, Front. Mol. Biosci., 04 September 2019

HSP70 OLIGOMERIZATION

Hsp70 Forms Oligomers by an Interaction

between the Interdomain Linker and the Substrate Binding Domains

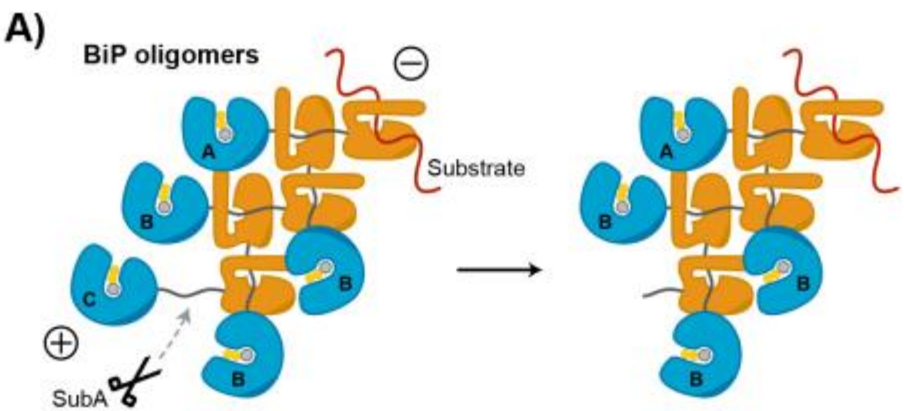


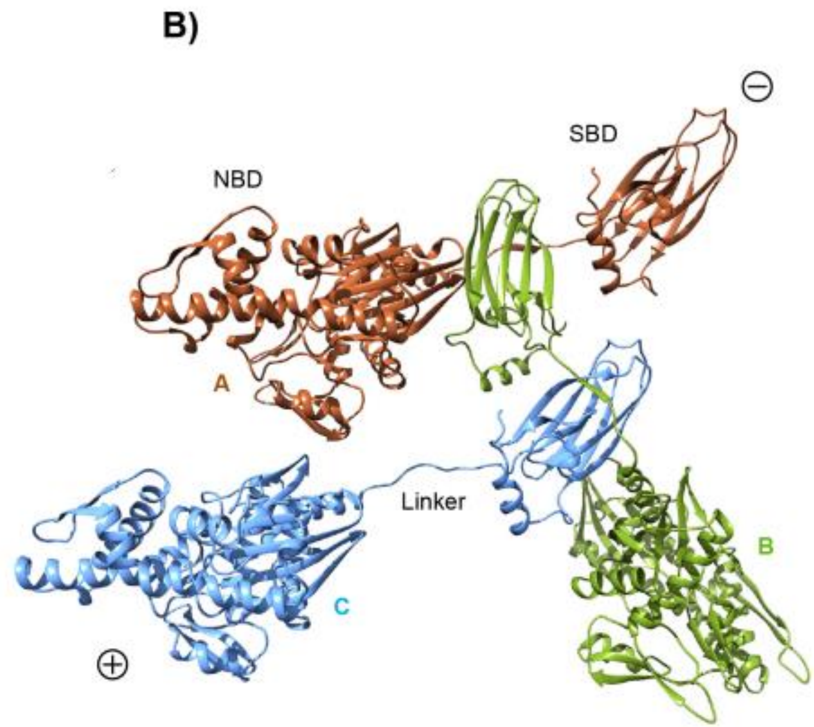
Hsp70 Oligomerization Is Mediated by an Interaction between the Interdomain Linker and the Substrate-Binding Domain, Cell Reports, Volume 11, Issue 5, 5 May 2015, Pages 759-769

HSP70 OLIGOMERIZATION

Hsp70 Forms Oligomers by an Interaction

between the Interdomain Linker and the Substrate Binding Domains





7a4u
7a4v

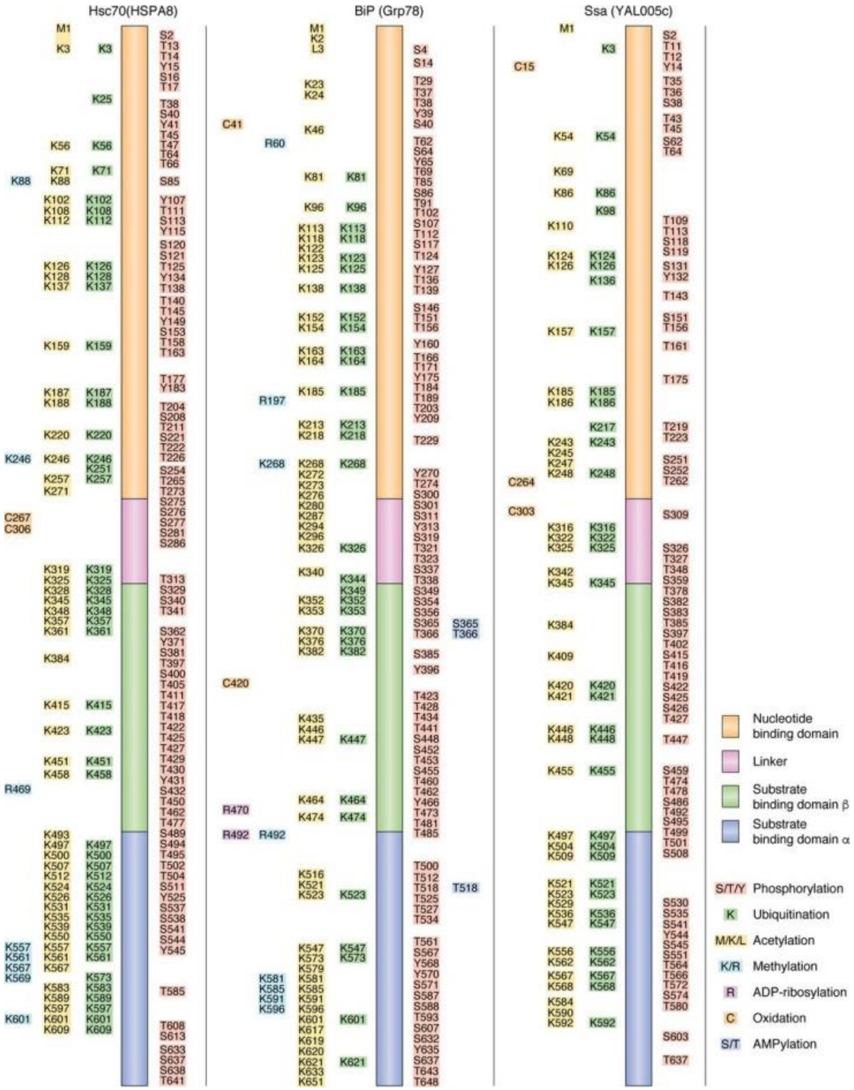
Steffen PreisslerClaudia RatoYahui YanLuke A PereraAron CzakoDavid Ron (2020)
Calcium depletion challenges endoplasmic reticulum proteostasis by destabilising BiP-substrate complexes eLife 9:e62601.

HSP70 PTMS CHAPERONE CODE

HSP70 REGULATION

PTMs Alter and Regulate Hsp70 Functions

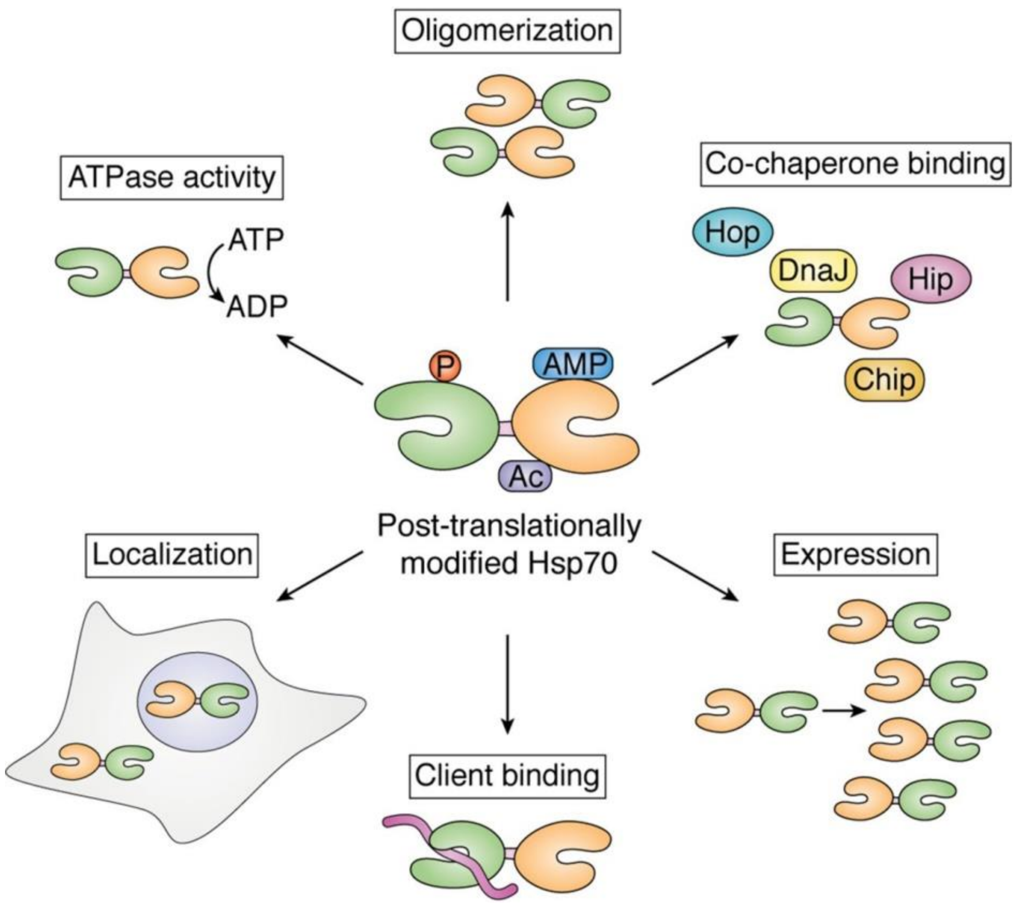
Post-translational modifications of Hsp70 family proteins: Expanding the chaperone code
JBC, Volume 295, Issue 31p10689-10708July 2020



HSP70 REGULATION

PTMs Alter and Regulate Hsp70 Functions

Post-translational modifications of Hsp70 family proteins: Expanding the chaperone code
JBC, Volume 295, Issue 31p10689-10708July 2020



THANKS FOR LISTENING